R. Weizmann KIENDREBEOGO, PhD

Postdoctoral Researcher · IRFU / CEA Paris-Saclay, Gif-sur-Yvette, France

weizmann.kiendrebeogo@oca.eu · ramodgwende-weizmann.kiendrebeogo@cea.fr

weizmannk.github.io · github.com/weizmannk · ORCID: 0000-0002-9108-5059 · Google Scholar

Research Interests

Multi-messenger Astronomy · Gravitational Waves · Time-Domain Astronomy · Kilonovae and Compact Binary Mergers · Observation Planning and Scheduling · Bayesian Inference · Machine Learning and Deep Learning for Astronomy

Education

PhD in Planetary and Universe Sciences, Université Côte d'Azur, Nice, France — September 2024

Joint supervision with Université Joseph Ki-Zerbo, Ouagadougou, Burkina Faso.

Thesis: *Developments for the observation and characterization of gravitational-wave multi-messenger sources during LIGO-Virgo-KAGRA observation campaigns*

M.Sc. (Master II) in Astrophysics & Materials — December 2020

Université Joseph Ki-Zerbo, Ouagadougou, Burkina Faso

B.Sc. in Fundamental Physics — October 2017

Université Nazi Boni, Bobo-Dioulasso, Burkina Faso

Two-year university degree in Mathematics, Physics & Computer Science (MPI) — 2016

Université Nazi Boni, Bobo-Dioulasso, Burkina Faso

Appointments

Postdoctoral Researcher, IRFU / CEA Paris-Saclay, France — January 2025 - present

Development and maintenance of Astro-COLIBRI, a platform for real-time multi-wavelength and multi-messenger transient analysis (ANR MOTS project). Multi-messenger follow-up strategies and low-latency inference for gravitational-wave counterparts.

Full-Time Physics and Chemistry Teacher (High School) — 2018 - 2021

Complexe Scolaire Sainte Marie Eugénie, Ouagadougou, Burkina Faso

Honors & Distinctions

Qualification for Associate Professor (Maître de conférences) — 2026

French National Council of Universities (CNU), Section 34 — Astrophysics

Research Grants

INSU (CNRS) Research Grant — AAP ASTRO + PNP 2026 — 2026

Principal Investigator – €5,900 (Priority P0)

Funds a 6-month M2 research internship.

Project: Optimized GW Detection and Multi-Messenger Follow-Up (Rubin Observatory, CTAO).

International co-supervision: IRFU/CEA, ARTEMIS (OCA), and Université Joseph Ki-Zerbo.

Fellowships & Awards

Gravitational Waves / Multi-Messenger Astronomy Axis Award — 2023 - 2024

Observatoire de la Côte d'Azur, Nice, France

Projet d'Appui à l'Enseignement Supérieur (PAES) — January 2023 - June 2023

Université Joseph Ki-Zerbo, Ouagadougou, Burkina Faso

Heising-Simons Foundation, grant 2021-2975 — June 2022 - August 2022

Las Cumbres Observatory

CNRS Sub-Saharan Africa Grant November 2021 - February 2022
Astrophysics and Planetary Sciences in West Africa (AWA)

Henri Poincaré Junior Program August 2020 - October 2020
Observatoire de la Côte d'Azur, Nice, France

Publications

*Author name in **bold**. Full list on **Google Scholar***

Submitted & In Preparation

1. **Kiendrebeogo** et al. (2026), "NMMA–Astro-COLIBRI: An Automated Light-Curve Classification Service for Supernovae in the Multi-Survey Era." *Submitted to The Astrophysical Journal Supplement Series.* https://doi.org/10.48550/arXiv.2608.17568
2. Schüssler et al. (2026), "AI-Assisted Extraction of Follow-up Observations from GCN Circulars in Astro-COLIBRI." *Submitted to Publications of the Astronomical Society of the Pacific.* https://doi.org/10.48550/arXiv.2608.23270
3. **Kiendrebeogo** et al. (2026), "Early UV Follow-up of Gravitational-Wave Events with the Ultraviolet Transient Astronomy Satellite (ULTRASAT)." *In preparation.*
4. **Kiendrebeogo** et al. (2026), "Refinements of Gravitational-Wave Observation Strategies Using O4a Performance." *In preparation.*

Refereed Publications

5. Johnson, Sravan, **Kiendrebeogo**, Coughlin, Davis, Toivonen, Jegou du Laz, Erb, Ahumada, Barna, Helou, Smith, Rusholme, Laher and Mahabal (2026), "Low-latency Forecasts of Kilonova Light Curves for Rubin and ZTF." *Publications of the Astronomical Society of the Pacific* **138**, 014103. https://doi.org/10.1088/1538-3873/ae2e87
6. **Kiendrebeogo** et al. (2025), "Application of Nonlinear Noise Regression in the Virgo Detector." *Physical Review D* **112**, 083053. https://doi.org/10.1103/hwnc-r7yy
7. Singer, Criswell, Leggio, **Kiendrebeogo**, Coughlin, Earnshaw, Gezari, Grefenstette, Harrison, Kasliwal, Morris, Tollerud and Cenko (2025), "Optimal Follow-up of Gravitational-wave Events with the UltraViolet EXplorer (UVEX)." *Publications of the Astronomical Society of the Pacific* **137**, 074501. https://doi.org/10.1088/1538-3873/adcfc6
8. Criswell, Leggio, Coughlin, Singer, **Kiendrebeogo**, Andreoni, Toivonen, Earnshaw, Gezari, Grefenstette, Harrison and Kasliwal (2025), "Electromagnetic Follow-up to Gravitational Wave Events with the UltraViolet EXplorer (UVEX)." *Publications of the Astronomical Society of the Pacific* **137**, 054101. https://doi.org/10.1088/1538-3873/adcd9c
9. Ahumada, . . . , **Kiendrebeogo**, . . . et al. (2024), "Searching for Gravitational Wave Optical Counterparts with the Zwicky Transient Facility: Summary of O4a." *Publications of the Astronomical Society of the Pacific* **136**, 114201. https://doi.org/10.1088/1538-3873/ad8265
10. Tosta e Melo, . . . , **Kiendrebeogo**, . . . et al. (2024), "Ready for O4 II: GRANDMA Observations of Swift GRBs over Eight Weeks in Spring 2022." *Astronomy & Astrophysics* **682**, A141. https://doi.org/10.1051/0004-6361/202347938
11. Andreoni, Coughlin, Criswell, Bulla, Toivonen, Singer, Palmese, Burns, Gezari, Kasliwal, **Kiendrebeogo**, Mahabal, Moriya, Rest, Scolnic, Simcoe, Soon, Stein and Travouillon (2024), "Enabling Kilonova Science with the Nancy Grace Roman Space Telescope." *Astroparticle Physics* **155**, 102904. https://doi.org/10.1016/j.astropartphys.2023.102904
12. **Kiendrebeogo** et al. (2023), "Updated Observing Scenarios and Multimessenger Implications for the International Gravitational-wave Networks O4 and O5." *The Astrophysical Journal* **958**, 158. https://doi.org/10.3847/1538-4357/acfcb1
13. Pang, Dietrich, Coughlin, Bulla, Tews, Almualla, Barna, **Kiendrebeogo**, Kunert, Mansingh, Reed, Sravan, Toivonen, Antier, Vanden Berg, Heinzel, Nedora, Salehi, Sharma, Somasundaram and Van Den Broeck (2023), "An Updated Nuclear-Physics and Multi-Messenger Astrophysics Framework for Binary Neutron Star Mergers." *Nature Communications* **14**, 8352. https://doi.org/10.1038/s41467-023-43932-6
14. Coughlin, . . . , **Kiendrebeogo**, . . . et al. (2023), "A Data Science Platform to Enable Time-Domain

Astronomy." *The Astrophysical Journal Supplement Series* **267**, 31. https://doi.org/10.3847/1538-4365/acdee1

15. Kunert, Antier, Nedora, Bulla, Pang, Anand, Coughlin, Tews, Barnes, Hussenot-Desenonges, Healy, Jegou du Laz, Pilloix, **Kiendrebeogo** and Dietrich (2023), "Bayesian Model Selection for GRB 211211A through Multiwavelength Analyses." *Monthly Notices of the Royal Astronomical Society* **527**, 3900–3911. https://doi.org/10.1093/mnras/stad3463
16. Kann, ..., **Kiendrebeogo**, ... et al. (2023), "GRANDMA and HXMT Observations of GRB 221009A: The Standard Luminosity Afterglow of a Hyperluminous Gamma-Ray Burst." *The Astrophysical Journal Letters* **948**, L12. https://doi.org/10.3847/2041-8213/acc8d0
17. Aivazyan, ..., **Kiendrebeogo**, ... et al. (2022), "GRANDMA Observations of ZTF/Fink Transients during Summer 2021." *Monthly Notices of the Royal Astronomical Society* **515**, 6007–6022. https://doi.org/10.1093/mnras/stac2054

Conference Proceedings

18. Ashkar, Seglar-Arroyo, Schüssler, **Kiendrebeogo** and de Bony de Lavergne (2025), "tilepy: Smart Scheduling for Multi-Messenger Astronomy from Earth to Orbit." Proceedings of the 39th International Cosmic Ray Conference (ICRC2025). https://doi.org/10.48550/arXiv.2508.07824
19. Schüssler, Cornejo, Costa, Jaroschewski and **Kiendrebeogo** (2025), "Astro-COLIBRI: Empowering Citizen Scientists in Multi-Messenger Astrophysics." https://doi.org/10.48550/arXiv.2507.10226
20. Schüssler, Cornejo, Costa, Jaroschewski and **Kiendrebeogo** (2025), "Astro-COLIBRI: A Comprehensive Platform for Real-Time Multi-Messenger Astrophysics." https://doi.org/10.48550/arXiv.2507.06616

Selected Collaboration Publications (LIGO–Virgo–KAGRA, GRANDMA)

21. LIGO Scientific, Virgo and KAGRA Collaborations (2025), "GWTC-4.0: An Introduction to Version 4.0 of the Gravitational-Wave Transient Catalog." https://arxiv.org/abs/2508.18080
22. LIGO Scientific, Virgo and KAGRA Collaborations (2025), "GWTC-4.0: Methods for Identifying and Characterizing Gravitational-wave Transients." https://arxiv.org/abs/2508.18081
23. LIGO Scientific, Virgo and KAGRA Collaborations (2025), "GWTC-4.0: Updating the Gravitational-Wave Transient Catalog with Observations from the First Part of the Fourth LIGO–Virgo–KAGRA Observing Run." https://arxiv.org/abs/2508.18082
24. LIGO Scientific, Virgo and KAGRA Collaborations (2025), "GWTC-4.0: Population Properties of Merging Compact Binaries." https://arxiv.org/abs/2508.18083
25. LIGO Scientific, Virgo and KAGRA Collaborations (2025), "All-sky Search for Long-duration Gravitational-wave Transients in the First Part of the Fourth LIGO–Virgo–KAGRA Observing Run." https://arxiv.org/abs/2507.12282
26. LIGO Scientific, Virgo and KAGRA Collaborations (2025), "Search for Continuous Gravitational Waves from Known Pulsars in the First Part of the Fourth LIGO–Virgo–KAGRA Observing Run." *The Astrophysical Journal* **983**, 99. https://doi.org/10.3847/1538-4357/adb3a0
27. LIGO Scientific, Virgo and KAGRA Collaborations (2025), "Swift-BAT GUANO Follow-up of Gravitational-wave Triggers in the Third LIGO–Virgo–KAGRA Observing Run." *The Astrophysical Journal* **980**, 207. https://doi.org/10.3847/1538-4357/ad9749
28. LIGO Scientific, Virgo and KAGRA Collaborations (2025), "Search for Gravitational Waves Emitted from SN 2023ixf." *The Astrophysical Journal* **985**, 183. https://doi.org/10.3847/1538-4357/adc681
29. LIGO Scientific, Virgo and KAGRA Collaborations (2024), "Observation of Gravitational Waves from the Coalescence of a 2.5–4.5 $M_\odot$ Compact Object and a Neutron Star." *The Astrophysical Journal Letters* **970**, L34. https://doi.org/10.3847/2041-8213/ad5beb
30. LIGO Scientific, Virgo and KAGRA Collaborations (2024), "Ultralight Vector Dark Matter Search Using Data from the KAGRA O3GK Run." *Physical Review D* **110**, 042001. https://doi.org/10.1103/PhysRevD.110.042001
31. LIGO Scientific, Virgo and KAGRA Collaborations (2024), "A Search Using GEO600 for Gravitational Waves Coincident with Fast Radio Bursts from SGR 1935+2154." *The Astrophysical Journal*

977, 255. https://doi.org/10.3847/1538-4357/ad8de0

32. Agayeva et al. (2022), "The GRANDMA Network in Preparation for the Fourth Gravitational-wave Observing Run." https://doi.org/10.1117/12.2630240

Technical Reports & Public Data Releases

1. **Kiendrebeogo**, Cornejo Avila, Bisero, Cellier, Ciric, Jaroschewski, Saint-Paul and Schüssler (2026), *NMMA Fits for SN 2021ugl (NMMA–Astro-COLIBRI paper)*. Zenodo, v1.0.0. https://doi.org/10.5281/zenodo.21771188
2. **Kiendrebeogo**, Farah, Talbot, Coughlin and Singer (2025), *Maximum-Posterior Hyperparameters of the Broken Power Law + 2 Peak Population Model from GWTC-4.0*. LIGO Document T2500311. https://dcc.ligo.org/LIGO-T2500311/public
3. **Kiendrebeogo**, Coughlin, Plestkova and Hoggard (2024), *Observing Scenarios Simulations for the HLVK Configuration for O4 Using 20 Million Injections*. Zenodo. https://doi.org/10.5281/zenodo.12693653
4. **Kiendrebeogo**, Toivonen, Coughlin and Dent (2023), *O4 Statistical Predictions with the PDB/GWTC-3 Distribution for HL and HLV*. LIGO Document T2300385. https://dcc.ligo.org/T2300385
5. **Kiendrebeogo**, Coughlin, Farah, Foley, Gray and Singer (2023), *Simulated Detection and Localization in LIGO's HL Configuration for O4a and HLVK for O4b and O5*. Zenodo. https://zenodo.org/records/10078926
6. Andrade, Coughlin, Noysena, **Kiendrebeogo** et al. (2022), *GRB 220325A: GRANDMA Observations*. GCN Circular 31804.
7. Coughlin, Farah, Foley, Gray, Singer and **Kiendrebeogo** (2022), *LIGO/Virgo/KAGRA Observing Capabilities: Simulated Detections and Localization for O4 and O5*. Zenodo. https://doi.org/10.5281/zenodo.7026209
8. **Kiendrebeogo**, Singer and Coughlin, *Observing Capabilities*. LIGO/Virgo/KAGRA Public Alerts User Guide. https://emfollow.docs.ligo.org/userguide/capabilities.html

Selected Open-Source Contributions

1. `NMMA-Astro-COLIBRI`: Reproducibility package for an automated supernova light-curve classification service, built on the NMMA nested-sampling framework; includes the fits and analysis scripts behind the SN 2021ugl case study. https://nmma.live/ ·https://github.com/astro-transients/nmma-astrocolibri-sn2021ugl
2. `Astro-COLIBRI`: Real-time multi-messenger astrophysics platform: a public API, cloud alert system, and website/mobile apps that filter and contextualize incoming alerts (GRBs, FRBs, supernovae, gravitational waves) so astronomers can react quickly to transient events. https://astro-colibri.com
3. `SkyPortal`: Collaborative astronomical data platform for time-domain astronomy, providing the infrastructure to manage, visualize, and analyze transient discoveries across a research team. https://github.com/skyportal/skyportal
4. `NMMA`: Python library for nuclear physics and multi-messenger inference, combining gravitational-wave, electromagnetic, and other data to constrain the physics of binary neutron star mergers. https://github.com/nuclear-multimessenger-astronomy/nmma
5. `M4OPT`: Open-source toolkit that jointly solves telescope tiling and scheduling via mixed-integer programming, coordinating space- and ground-based observatories for rapid follow-up of gravitational-wave and neutrino events. https://github.com/m4opt/m4opt
6. `tilepy`: Python package for gravitational-wave follow-up scheduling with Imaging Atmospheric Cherenkov Telescopes (IACTs), tiling the sky-localization region to optimize observation strategy when GW events are poorly localized. https://github.com/astro-transients/tilepy
7. `GWEMOPT`: Tool for optimizing electromagnetic follow-up of gravitational- wave events, coordinating multi-telescope tiling searches to efficiently locate optical/IR counterparts. https://github.com/skyportal/gwemopt
8. `ligo.skymap`: Toolkit for reading, writing, generating, and visualizing gravitational-wave probability sky maps, including the BAYESTAR rapid-localization algorithm and 3D sky map analysis. https:

//lscsoft.docs.ligo.org/ligo.skymap/

9. `observing-scenarios-simulations`: Scripts and data generating LIGO/Virgo/KAGRA compact-binary- coalescence observing-scenario predictions, simulating source populations and sky localization with BAYESTAR. https://github.com/lpsinger/observing-scenarios-simulations
10. `DeepClean / Virgo-DeepClean`: 1D convolutional neural network for non-linear, non-stationary noise subtraction in GW data. https://github.com/weizmannk/Virgo-DeepClean
11. `cbc-population-distributions`: Tools to draw compact-binary-coalescence populations from the GWTC-4.0 model (FullPop-4.0 component-mass distribution, power-law redshift distribution) for large-scale population sampling and visualization. https://cbc-population-distributions.readthedocs.io/en/latest/
12. `EarthOrbitPlan`: Educational framework walking students and researchers through an end-to-end compact-binary-coalescence workflow, from gravitational-wave detection to planning electromagnetic follow-up campaigns across multiple telescopes. https://earthorbitplan.readthedocs.io/en/latest/

Selected Talks

1. **AI for Gravitational Waves Workshop, CERN, Geneva, Switzerland** May 2026
   Contributed talk: Deep Learning Approaches to Gravitational-Wave Detector Noise Subtraction

2. **Journée de l'ATPEM 2025 (PNHE), Paris, France** October 2025
   Contributed talk: Early UV Follow-up of Gravitational-Wave Events with ULTRASAT

3. **4th Astro-COLIBRI Multi-Messenger Astrophysics Workshop, Institut Pascal, Paris-Saclay, France** October 2025
   Invited speaker: M4OPT: Multi-Mission Multi-Messenger Observation Planning Toolkit

4. **ACME Conference on Gravitational Waves and Multi-Messenger Astronomy, Observatoire Midi-Pyrénées, Toulouse, France** April 2025
   Contributed talk: Application of Non-Linear Noise Regression in the Virgo Detector

5. **LVK Cosmology Working Group, internal seminar, online** May 2023
   Invited group talk: Updated Observing Scenarios and H0 Projections for O4 and O5

Teaching & Supervision

Principal Investigator and co-supervisor, M2 research internship (6 months) 2026
Funded by the INSU (CNRS) AAP ASTRO + PNP grant; co-supervision across IRFU/CEA, ARTEMIS (OCA), and Université Joseph Ki-Zerbo.

Mathematics Teacher (44 hours), Department of Mathematics, Université Côte d'Azur, Nice, France
2023 - 2024

Research Project Co-Supervisor, School of Physics and Astronomy, University of Minnesota, USA
2022 - 2023
Co-supervised one undergraduate (B.Sc., 2022) and one master's (M.Sc., 2023) research project.

Physics and Chemistry Teacher (High School), full-time, Complexe Scolaire Sainte Marie Eugénie, Ouagadougou, Burkina Faso 2018 - 2021

Service to the Community

Referee, Science and Technology Review, CNRST, Burkina Faso January 2026

Member, LIGO-Virgo-KAGRA Collaboration

Member, GRANDMA Collaboration

Maintainer, NMMA, Astro-COLIBRI, tilepy, M4OPT, GWEMOPT

Contributor, Observing Capabilities section of the LVK Public Alerts User Guide — the reference resource used by the electromagnetic follow-up community.

Schools, Workshops & Research Visits

1. **2nd MaNiTou International Summer School on Gravitational Waves** 3 July 2023 – 8 July 2023
   Observatoire de la Côte d'Azur, Université Côte d'Azur, Nice, France
2. **Research visit — collaboration on the Vera C. Rubin Observatory (LSST) project** June 2022 – September 2022
   School of Physics and Astronomy, University of Minnesota, USA
3. **Internship — gravitational waves from EMRIs and the stochastic background** August 2020 – October 2020
   Observatoire de la Côte d'Azur, Nice, France
4. **Internship — imaging differences for M87* modelled as a rotating black hole vs. a rotating wormhole** January 2020 – March 2020
   Observatoire de Paris, Meudon, France
5. **WAISSYA (West African Summer School for Young Astronomers), postgraduate stream** November 2019
   Obasanjo Space Centre, Abuja, Nigeria

Technical Skills

Programming: Python (advanced), C/C++, Bash, Dart / Flutter (Astro-COLIBRI mobile app), LaTeX

Scientific computing: NumPy, SciPy, pandas, Astropy, Matplotlib, CuPy (GPU-accelerated arrays), HEALPix / ligo.skymap

Machine learning: PyTorch, TensorFlow, scikit-learn, Deep learning for time series (CNN)

Bayesian inference: bilby, NMMA, Nested sampling, MCMC, Population inference

Software engineering: Git (GitHub, GitLab), Docker, pytest, PyPI/conda, CI/CD, REST APIs, Technical documentation (Sphinx, MkDocs, Read the Docs)

High-performance computing: HTCondor and SLURM clusters, Large-scale simulation campaigns (>20M injections)

Languages

French (native) · English (fluent, working language) · Mooré (native)